Pre-Analysis Plan: Many Daughters: Impact of Having Daughters on Attitudes, Preferences, and Behaviors

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Abstract

This pre-analysis plan is part of the #ManyDaughters project, a coordinated global initiative[1] in which independent research teams test a shared set of hypotheses on whether having daughters influences individuals' attitudes, preferences, and behaviors. The project uses the German Socio-Economic Panel (SOEP-Core, 2000–2020) as a common data source, and participating teams submit pre-analysis plans (PAP) and analysis code.

This study tests four pre-specified hypotheses about the effects of having daughters. Specifically, we examine whether having daughters (H1) increases support for gender equality (measured by attitudes), (H2) leads to more left-leaning political preferences, (H3) promotes greater gender equality in the household (measured by behaviors or outcomes), and (H4) has a stronger positive effect on support for gender equality among men than among women.

We conduct preliminary analyses using a 5% random subsample provided by the project coordinators. This pre-analysis plan outlines the hypotheses, variable construction, analytical strategy, and reports preliminary results based on the limited data available, following a brief review of related literature and methods.

[1]`https://www.manydaughters.com/`

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1 Background and Motivation

Do children influence their parents' political and social preferences? A growing body of research suggests that the gender of one's children, in particular, can shape adult beliefs and behaviors. Several studies have examined whether having daughters increases support for gender equality across different domains—including political preferences (Washington, 2008), judicial decisions (Glynn and Sen, 2015), and household and workplace attitudes (Ronchi et al., 2021). Oswald and Powdthavee (2010) find that having daughters is associated with more left-leaning political views, while Shafer and Malhotra (2011) show that a daughter reduces support for traditional gender roles among men. Moreover, Borrell-Porta, Costa-Font, and Philipp (2019) document the so-called "mighty girl" effect, finding that parenting school-aged daughters significantly decreases fathers' endorsement of traditional male breadwinner norms. Together, these findings highlight the potential for child gender to shape adult attitudes and inform broader social and political dynamics.

This study contributes to the growing literature by examining pre-specified hypotheses using data from SOEP-Core, a large and representative longitudinal panel of the German population. The richness of the dataset—in terms of both the range of gender-related attitudinal, behavioral, and preference measures, and its inclusion of younger cohorts—offers an opportunity to revisit earlier findings and explore heterogeneous effects across demographic groups. While the full scope of this crowd science initiative will be shaped by cross-team analyses, the project has the potential to yield cumulative insights through coordinated hypothesis testing and meta-analytic synthesis.

2 Methods

2.1 Data Source

We use data from the German Socio-Economic Panel (SOEP-Core)[2], a nationally representative longitudinal survey of private households in Germany that began in 1984. For this study, we use a project-specific dataset covering the years 2000 to 2020, which includes all individuals who completed the individual questionnaire in at least one of those years (n = 90,357). The dataset provides rich information on respondents' socio-demographic characteristics, family structure, child gender composition, political preferences, and attitudes toward gender roles and equality.

[2]Goebel et al., 2019; Siegers, Steinhauer, and Schütt, 2022

2.2 Sample and Inclusion Criteria

The unit of analysis is the individual respondent who completed at least one individual questionnaire between 2000 and 2020. Respondents are included in hypothesis-specific models if they have valid (non-missing) values on the relevant dependent, independent, and control variables.

To identify whether a respondent has at least one daughter, we construct a binary treatment variable, `has_daughter`, based on the SOEP variables `kidsex01` through `kidsex19`, which indicate the gender of each biological child. This approach allows us to capture daughter presence more comprehensively than relying on `pld0026` (number of daughters), which is missing for the vast majority of respondents.

A methodological consideration arises regarding the control group—i.e., individuals coded as `has_daughter = 0`. One possibility is to restrict the sample to parents only and compare respondents with daughters to those with only sons. While this would facilitate stronger causal identification, it substantially reduces sample size and diverges from the broad language of the pre-specified hypotheses.

Accordingly, we follow the broader interpretation and adopt a full-sample approach: respondents are coded as 1 if they have at least one daughter, and 0 otherwise (including both respondents with only sons and those with no children). This decision aligns with the project guidelines and allows for broader inclusion.

Inclusion criteria are applied separately for each hypothesis. Respondents are retained only if they have valid responses (i.e., not coded as –8, –5, –2, or –1) to the relevant dependent variable:

- `plj0620_modified` for H1 and H4,
- `plm0644_modified` for H2,
- `plh0303_modified` for H3.

We also apply listwise deletion for the control variables. Additionally, for H4, respondents must have non-missing values on both `male` and the interaction term `has_daughter` × `male`.

While the SOEP includes tens of thousands of individuals, valid response counts for our dependent variables are substantially smaller. Furthermore, our use of listwise deletion for missing control variables is expected to further reduce the effective sample size. As a result, we do not formally assess statistical power at this stage. However, if the final analytic sample sizes fall below commonly accepted thresholds for minimum detectable effect sizes (e.g., to achieve 80% power at a 5% significance level), the specification of our outcome, predictor, or control variables may need to be revised accordingly.

2.3 Variable Construction

All variables are constructed using standard SOEP naming conventions. For each hypothesis, we define the dependent, independent (treatment), and control variables based on available items from the relevant SOEP-Core datasets, ensuring consistency with the project's pre-specified analytical framework.

2.3.1 Dependent variables

- **H1:** Support for gender equality, operationalized using the SOEP survey item *"Women have the same rights as men"* (variable name: `plj0620`). Responses are given on an 11-point scale (0 = "Should definitely not happen in a democracy", 10 = "Should definitely happen in a democracy"). Invalid responses (i.e., -8 = "Question not part of survey this year", -5 = "Not included in this version", and -1 = "No answer") are recoded as missing. The cleaned variable is named `plj0620_modified` and observations with missing values on this variable are dropped from the sample.

- **H2:** Left-leaning political preferences are associated with support for redistributive fairness, as measured by the SOEP item: *"A society is fair when income and wealth are equally distributed among all people"* (variable: `plm0644`). Respondents answered on a 5-point Likert scale, where 1 indicates strong agreement and 5 indicates strong disagreement. To align with the hypothesis (i.e., that stronger support for redistribution reflects more left-leaning preferences), this variable is reversed by multiplying responses by -1. Additionally, invalid responses (-8, -5, -1) are recoded as missing. The final cleaned and reversed variable used in the analysis is named `plm0644_modified`.

- **H3:** Support for gender equality in the household, measured via agreement with the SOEP item *"Men should be involved in housework just as much as women."* (`plh0303`). Responses are on a scale from 1 (strong agreement) to 4 (strong disagreement), and the variable is reverse-coded so that higher values reflect greater support for gender equality. Invalid responses (i.e., -8, -5, -1) are recoded as missing. The cleaned and reverse-coded variable used in analysis is labeled `plh0303_modified`.

- **H4:** Support for gender equality as measured by the survey item *"Women have the same rights as men"* (`plj0620`). This variable is modified to exclude invalid responses (-8, -5, -1). The cleaned version used in the interaction model is also `plj0620_modified`.

These variables are based on SOEP survey items capturing respondents' views and opinions, cleaned and reverse-coded where appropriate.

2.3.2 Treatment variable

A binary indicator `has_daughter`, equal to 1 if the respondent has at least one daughter (constructed from the child gender variables `kidsex01` to `kidsex19`, where `kidsex` = 2 indicates female), and 0 otherwise.

2.3.3 Control variables

To reduce omitted variable bias, we include the following controls. All transformations match SOEP variable definitions, and invalid or missing values are excluded as described below:

- `male`: A binary variable equal to 1 if respondent is male (`sex` = 1), 0 if female (`sex` = 2). Other values are set to missing. We control for respondent gender, which is strongly associated with baseline levels of support for gender equality.
- `age`: Derived from `bioage`, excluding values equal to -2, which indicate missing or invalid data. We include age to account for generational differences in social and political attitudes.
- `years_of_education`: Based on `pgbilzeit`; values of -1 or below are excluded. Education is associated with political ideology and gender attitudes, and may confound the relationship between parenthood and beliefs.
- `household_income`: Based on `hghinc`; only values ≥ 0 are retained. Income may shape redistributive preferences and views on gender roles.
- `marital_status`: Taken from `pgfamstd`, with only values > 0 retained. Used as a categorical variable. Marital status may influence exposure to traditional or egalitarian household dynamics.
- `employment_status`: Taken from `pgemplst`, where only values $\neq -2$ are included. Used as a categorical variable. Employment status may affect views on work-family balance and gender roles.
- `migrant`: Based directly on `migback`; used as-is without recoding. Migration background may be associated with different cultural norms regarding gender and family roles.

2.4 Estimation Strategy

We estimate ordinary least squares (OLS) regression models to test each hypothesis. Robust standard errors are used in all regressions. Main specifications are estimated without weights due to the fact that only 5% of the data is used in this analysis. All reported p-values are unadjusted. Given that the hypotheses were pre-specified and each model tests a conceptually distinct hypothesis, we do not apply multiple hypothesis correction at this stage.

2.5 Use of 5% Sample

In accordance with the guidelines of the #ManyDaughters project, we have written and tested our analysis code using a 5% random subsample of the full dataset provided to participating research teams. This subset was drawn at the household level and is intended to support teams in preparing and validating their pre-analysis plans. All analyses using this subsample were conducted prior to the release of the full dataset.

The regression models described in this pre-analysis plan were executed on the 5% sample, and we report preliminary results from these analyses. These results are intended to illustrate how our analytical strategy is implemented in practice; however, all findings presented here should be interpreted as exploratory until confirmed on the full dataset. The construction of key variables—such as `has_daughter`—as well as decisions regarding sample inclusion and model structure, were guided primarily by the original hypothesis statements and the structure of the data, rather than by the observed results.

Accordingly, this pre-analysis plan remains confirmatory in nature. Our access is limited to the 5% subsample. The final analysis will be conducted by the project coordinators using the full dataset and the analysis code we submit.

3 Hypothesis Tests and Preliminary Results

3.1 Hypotheses

We test the following four hypotheses in this study:

H1. Having daughters leads to stronger support for gender equality.

H2. Having daughters leads to more left-leaning political preferences.

H3. Having daughters leads to more gender equality in the household.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women.

3.2 Variables

Section 2.3 provides a detailed description of the exact construction of all variables used in the analysis. Unless otherwise noted, invalid responses (e.g., –8 = "Question not part of survey this year", –5 = "Not included in this version", –2 = "Does not apply", and –1 = "No answer") are recoded as missing. To test the four hypotheses outlined above, this section lists all dependent and independent variables used in the analyses, along with the set of control variables. Each hypothesis has its own dependent variable, defined based on specific SOEP survey items. In contrast, the main independent variable (`has_daughter`) and the set of control variables are shared across all models.

3.2.1 Independent variable

We constructed a binary variable (`has_daughter`) indicating whether the respondent has at least one daughter, based on the variables `kidsex01` to `kidsex19`, which record the sex of up to 19 biological children. The independent variable `has_daughter` equal to 1 if the respondent has at least one daughter, and 0 otherwise. This approach allows us to capture daughter presence more comprehensively than relying on the SOEP variable `pld0026` (number of daughters), which contains substantial missing data and is not consistently collected across waves.

3.2.2 Dependent variables

- **H1 – Gender equality attitudes:** `plj0620_modified`
- **H2 – Left-leaning political preferences:** `plm0644_modified`
- **H3 – Gender equality in the household:** `plh0303_modified`
- **H4 – Gender equality attitudes by respondent gender:** `plj0620_modified`

3.2.3 Control variables

All models include the following covariates to adjust for relevant observable characteristics: `male`, `age`, `years_of_education`, `household_income`, `marital_status`, `employment_status`, and `migrant`. The construction of these control variables based on the SOEP dataset is described in detail in Section 2.3.

3.2.4 Interaction term

To test whether the effect of having daughters differs by respondent gender, we construct an interaction term between `has_daughter` and `male`.

$$\texttt{interaction} = \texttt{has_daughter} \times \texttt{male}$$

This interaction term allows us to test whether the coefficient on `has_daughter` varies between male and female respondents. It is only used in the analysis for Hypothesis 4.

3.3 Exact Analysis

We estimate separate ordinary least squares (OLS) regression models for each hypothesis. All models include robust standard errors to correct for potential heteroskedasticity and control for a common set of covariates as outlined previously. Model specifications are as follows:

H1: Daughters and Support for Gender Equality (Attitudes)

$$\texttt{plj0620_modified}_i = \beta_0 + \beta_1 \, \texttt{has_daughter}_i + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i$$

where $\texttt{plj0620_modified}_i$ denotes agreement with the statement that women have the same rights as men, and $\mathbf{X}_i$ is the vector of control variables.

H2: Daughters and Left-Leaning Political Preferences

$$\texttt{plm0644_modified}_i = \beta_0 + \beta_1 \, \texttt{has_daughter}_i + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i$$

where $\texttt{plm0644_modified}_i$ is the reversed scale of agreement with the statement that income and wealth are equally distributed among all people in a fair society.

H3: Daughters and Gender Roles in the Household

$$\texttt{plh0303_modified}_i = \beta_0 + \beta_1 \, \texttt{has_daughter}_i + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i$$

where $\texttt{plh0303_modified}_i$ captures reverse-coded agreement with the statement that men should be involved in housework just as much as women.

H4: Daughters and Gender Equality Attitudes by Respondent Gender

$$\texttt{plj0620_modified}_i = \beta_0 + \beta_1 \texttt{has_daughter}_i + \beta_2(\texttt{has_daughter}_i \times \texttt{male}_i) + \boldsymbol{\gamma}'\mathbf{X}_i + \varepsilon_i$$

This model tests whether the association between having daughters and support for gender equality (as in H1) differs by gender. Both the main effect of having daughters (β_1) and the interaction term with respondent gender (β_2) are of substantive interest in this specification.

In all models, robust standard errors are used. The null hypothesis $\beta_1 \leq 0$ (and $\beta_2 \leq 0$ in H4) is tested using one-tailed t-tests.

3.4 Preliminary Results

To validate our analysis pipeline and provide a preliminary look at the estimated effects, we executed each model on a randomly selected 5% subsample of the full dataset. The table below reports regression coefficients, robust standard errors, test statistics, p-values, and sample sizes for each hypothesis. These estimates are intended to illustrate the implementation of the pre-specified models and do not reflect final results; therefore, no definitive conclusions should be drawn regarding the hypotheses based on this subset.

Table 1: Preliminary Regression Results for H1–H4 (5% Sample)

Hypothesis	**Dependent Var.**	β_1	**SE**(β_1)	**t**(β_1)	**p**(β_1)	β_2	**SE**(β_2)	**t**(β_2)	**p**(β_2)	**Obs.**
H1	plj0620_modified	-0.3703	0.1914	-1.93	0.054	–	–	–	–	304
H2	plm0644_modified	0.3691	0.3214	1.15	0.253	–	–	–	–	147
H3	plh0303_modified	-0.0137	0.0867	-0.16	0.875	–	–	–	–	328
H4	plj0620_modified	-0.4134	0.3037	-1.36	0.175	0.0807	0.3945	0.20	0.838	304

Note: For each coefficient, we report the estimated value (β), its robust standard error, the corresponding t-statistic, and the p-value. H4 includes an interaction term between `has_daughter` and `male`; statistics for the interaction coefficient (β_2) are only reported for this model. All models include the full set of control variables listed in Section 3.2 ('Variables') and use robust standard errors.

3.5 Interpretation

Across all four hypotheses, our main parameter of interest is β_1, which is the regression coefficient on `has_daughter`.

We will consider the sign and significance of this coefficient in determining whether the results support each hypothesis. Specifically, a coefficient that is in the expected direction

and statistically significant at the 5% level ($p < 0.05$) will be interpreted as consistent with the corresponding hypothesis.

We test H4 using two separate one-tailed t-tests:

1. $H_0 : \beta_1 \leq 0 \quad \text{vs} \quad H_1 : \beta_1 > 0$
2. $H_0 : \beta_2 \leq 0 \quad \text{vs} \quad H_1 : \beta_2 > 0$

We conclude that H4 is supported only if both tests reject their respective null hypotheses at the 5% significance level.

We report point estimates, robust standard errors, and p-values for each model. The preliminary results obtained from the 5% random subsample allow for a basic assessment of the direction and potential statistical significance of each hypothesized relationship. These early findings are not intended for inference but are presented here to illustrate model outputs and ensure transparency in analytical procedures.

- **H1 (Gender Equality Attitudes):** The coefficient on `has_daughter` is negative and marginally significant ($p = 0.054$). Contrary to expectations, this suggests a slight decrease in support for gender equality among respondents with daughters. However, the estimate is imprecise and should be interpreted with caution given the limited sample.

- **H2 (Left-Leaning Political Preferences):** The coefficient is positive, as expected, but statistically insignificant ($p = 0.253$). While the direction is consistent with the hypothesis, the magnitude and uncertainty suggest no reliable relationship in this subsample.

- **H3 (Gender Equality in the Household):** The estimated coefficient is close to zero and highly insignificant ($p = 0.875$). This offers no preliminary evidence that having daughters is associated with more egalitarian views regarding men's roles in housework.

- **H4 (Gender Equality Attitudes by Respondent Gender):** The interaction coefficient (β_2) between `has_daughter` and `male` is positive but far from statistically significant ($p = 0.838$). The main effect of `has_daughter` remains negative and also statistically insignificant ($p = 0.175$). These findings do not support the hypothesis that the effect of having daughters on gender equality attitudes is stronger among men than women.

In sum, none of the four hypotheses receive strong support in the preliminary analysis using the 5% sample. These findings underscore the importance of full-sample estimation before drawing any conclusions. All hypotheses, model specifications, and variable constructions were finalized before conducting the preliminary analysis. The results reported here were not used to revise any aspect of the analysis plan.

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